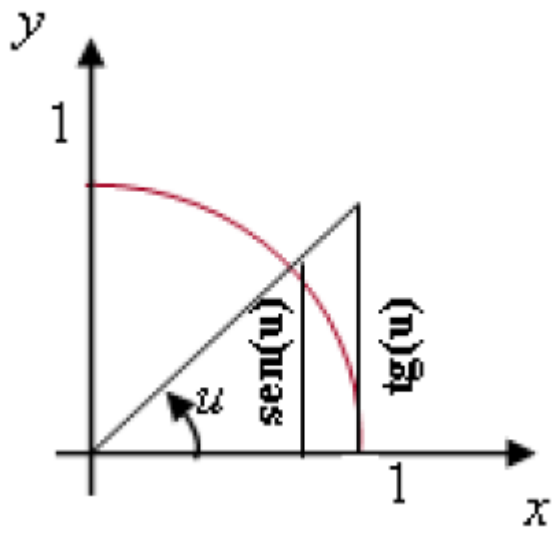
Limites Notáveis

Teorema 1: $\lim_{u\to 0}\dfrac{\operatorname{sen}(u)}{u}=1.$

Demonstração:



Comparando os comprimentos, temos que:

$$\operatorname{sen}(u) < u < tg(u)$$

$$\Rightarrow \quad \operatorname{sen} u < u < \frac{\operatorname{sen}(u)}{\cos(u)}$$

Sabemos que $\operatorname{sen}(u) > 0$ *e* $\cos(u) > 0$ *para* $u \in \left(0, \frac{\pi}{2}\right)$. *Assim, dividindo tudo por* $\operatorname{sen}(u)$, *obtém-se que:*

$$1 < \frac{u}{\operatorname{sen}(u)} < \frac{1}{\cos(u)}$$

$$\Rightarrow \quad 1 > \frac{\operatorname{sen}(u)}{u} > \cos(u)$$

$$\Rightarrow \quad \lim_{u\to 0} 1 < \lim_{u\to 0}\frac{\operatorname{sen}(u)}{u} < \lim_{u\to 0}\cos(u) \quad \Rightarrow \quad 1 < \lim_{u\to 0}\frac{\operatorname{sen}(u)}{u} < 1$$

$$\lim_{u\to 0}\frac{\operatorname{sen}(u)}{u}=1$$

Teorema 2: $\lim_{u\to 0} \dfrac{1-\cos(u)}{u} = 0.$

Demonstração:

$$L = \lim_{u\to 0} \frac{1-\cos(u)}{u} \cdot \frac{1+\cos(u)}{1+\cos(u)}$$

$$L = \lim_{u\to 0} \frac{1-cos^2(u)}{u(1+\cos(u))}$$

$$L = \lim_{u\to 0} \frac{sen^2(u)}{u(1+\cos(u))}$$

$$L = \lim_{u\to 0} \left(\frac{sen(u)}{u} \cdot \frac{sen(u)}{1+\cos(u)} \right)$$

$$L = \underbrace{\lim_{u\to 0} \left(\frac{sen(u)}{u} \right)}_{1} \underbrace{\lim_{u\to 0} \left(\frac{sen\,(u)}{1+\cos(u)} \right)}_{\frac{0}{2}} \quad \Rightarrow \quad \boxed{L = \lim_{u\to 0} \frac{1-\cos(u)}{u} = 0}$$

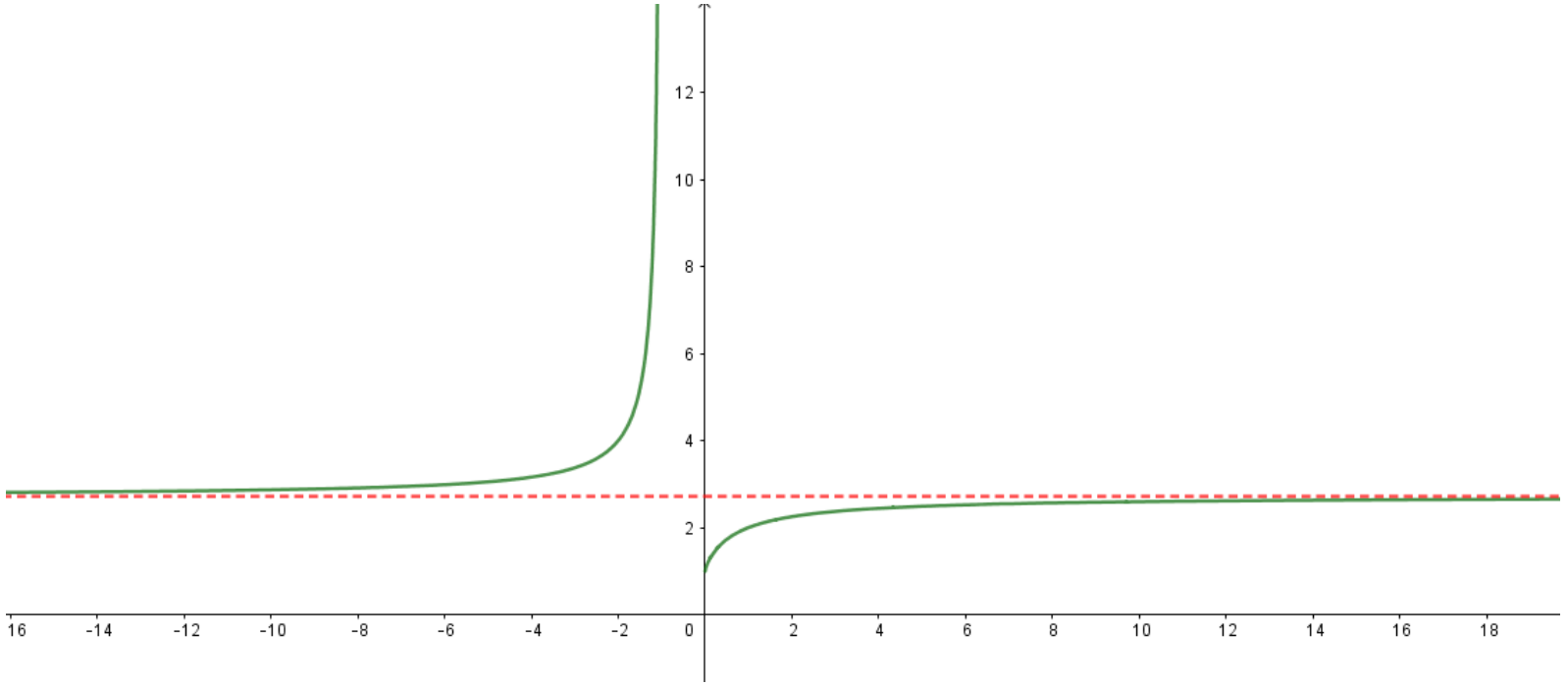
Teorema 3: $\lim_{u\to\pm\infty}\left(1+\frac{1}{u}\right)^u = e$

u	1	10	100	1000	10000	100000	1000000
(1+1/u)^u	2	2,59374246	2,704814	2,716924	2,718146	2,718268	2,71828

$$\lim_{u\to+\infty}\left(1+\frac{1}{u}\right)^u = e$$

u	-1	-10	-100	-1000	-10000	-100000	-1000000
(1+1/u)^u	x	2,867971991	2,731999	2,719642	2,718418	2,718295	2,718283

$$\lim_{u\to-\infty}\left(1+\frac{1}{u}\right)^u = e$$



Teorema 4: $\lim_{x\to 0}\dfrac{a^x-1}{x}=\ln a$, para $a>0$ e $a\neq 1$.

Definindo $u=a^x-1$.

Aplicando logaritmo neperiano em ambos os lados, temos que:

$$\ln(u+1)=\ln a^x \quad\Rightarrow\quad x=\frac{\ln(u+1)}{\ln a}.$$

Se $x\to 0$, *então* $u\to 0$.

$$L=\lim_{x\to 0}\frac{a^x-1}{x}=\lim_{u\to 0}\frac{u}{\frac{\ln(u+1)}{\ln a}}=\ln a.\lim_{u\to 0}\frac{u}{\ln(u+1)}=\ln a.\lim_{u\to 0}\frac{1}{\frac{\ln(u+1)}{u}}$$

$$\Rightarrow L=\ln a.\frac{1}{\lim\limits_{u\to 0}\left(\frac{\ln(u+1)}{u}\right)}=\ln a.\frac{1}{\ln\left(\lim\limits_{u\to 0}(u+1)^{\frac{1}{u}}\right)}=\ln a.\frac{1}{\ln e}=\ln a.$$

Exemplos:

Use os limites notáveis ou fundamentais, sempre que possível para calcular os limites dados a seguir.

a. $\lim\limits_{x\to 0} \dfrac{sen(3x)}{sen(5x)} = \dfrac{0}{0}$

$$L = \lim_{x\to 0} \frac{sen(3x)}{sen(5x)} = \lim_{x\to 0} \frac{sen(3x)}{sen(5x)} \cdot \frac{\frac{1}{x}}{\frac{1}{x}} = \lim_{x\to 0} \frac{\frac{sen(3x)}{x}}{\frac{sen(5x)}{x}} = \lim_{x\to 0} \frac{\frac{3\ sen(3x)}{3x}}{\frac{5\ sen(5x)}{5x}} = \frac{\lim\limits_{x\to 0} \frac{3\ sen(3x)}{3x}}{\lim\limits_{x\to 0} \frac{5\ sen(5x)}{5x}}$$

$$L = \frac{\lim\limits_{x\to 0} 3 \ . \lim\limits_{x\to 0} \frac{sen(3x)}{3x}}{\lim\limits_{x\to 0} 5 \ . \lim\limits_{x\to 0} \frac{sen(5x)}{5x}} = \frac{3 \lim\limits_{x\to 0} \frac{sen(3x)}{3x}}{5 \lim\limits_{x\to 0} \frac{sen(5x)}{5x}}$$

Definindo $u = 3x$ e $v = 5x$, temos que:

$$L = \frac{3 \lim\limits_{u\to 0} \frac{sen(u)}{u}}{5 \lim\limits_{v\to 0} \frac{sen(v)}{v}} \quad \Rightarrow \quad \boxed{L = \frac{3}{5}}$$